DESIGN AND DEVELOP A REACT JS BASED WEB APPLICATION FOR TICKET BOOKING OF AN INTERNAL DEPARTMENT EVENT

School of Computing

Bachelor of Technology
in
Computer Science & Engineering

By

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It is certified that the work contained in the Full Stack Application Development report titled ”Design and Develop a React JS Based Web Application For Ticket Booking of an Internal Department Event” by Mani Manjunath V(VTU24573) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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ABSTRACT

The digital transformation of educational institutions has streamlined many administrative tasks, yet the management of extracurricular activities, technical symposiums, and cultural fests often remains highly fragmented. Traditionally, campus events rely on physical ticket sales, disjointed Google Forms, and cash transactions, leading to long queues, mismanaged funds, and manual data entry errors [1]. The Campus Cultural platform is a purpose-built, full-stack web application designed specifically to solve these challenges within a college ecosystem. By providing a centralized digital hub, it allows event organizers to broadcast upcoming activities while empowering students to effortlessly discover, book, and access events using modern technologies [2].

Keywords: React JS, Event Management, Ticket Booking, Full Stack Development, Node.js, MySQL.

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
CSS	Cascading Style Sheets
CSV	Comma-Separated Values
DBMS	Database Management System
JWT	JSON Web Token
OAuth	Open Authorization
PDF	Portable Document Format
QR	Quick Response
SMTP	Simple Mail Transfer Protocol
SQL	Structured Query Language
UI/UX	User Interface / User Experience

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Chapter 1

INTRODUCTION

1.1 Introduction

The digital transformation of educational institutions has streamlined many administrative tasks, yet the management of extracurricular activities, technical symposiums, and cultural fests often remains highly fragmented. Traditionally, campus events rely on physical ticket sales, disjointed Google Forms, and cash transactions, leading to long queues, mismanaged funds, and manual data entry errors [3]. The Campus Cultural platform is a purpose-built, full-stack web application designed specifically to solve these challenges within a college ecosystem. By providing a centralized digital hub, it allows event organizers to broadcast upcoming activities while empowering students to effortlessly discover, book, and access events using modern technologies [4].

1.2 Aim of the Project

The primary aim of the Campus Cultural project is to architect an intuitive, scalable, and highly secure digital event reservation ecosystem [5]. Specifically, the project aims to eliminate manual processes, enhance security and prevent fraud, provide real-time analytics, and ensure seamless user experience.

1.3 Scope of the Project

The scope of this project encompasses the complete lifecycle of event management and ticketing. It includes the development of a student-facing portal for browsing event catalogs, managing personal academic profiles (including VTU numbers and department details), and processing secure payments [6]. For administrators, the scope includes an interface for creating and managing events, viewing comprehensive booking histories, exporting data to CSV for offline use, and a dedicated QR scanning module for gate validation.

The scope explicitly excludes non-campus related public event hosting, physical hardware integration for turnstiles, and direct accounting/tax generation, focusing strictly on the reservation, payment, and digital entry aspects of campus fests [7].

1.4 Framework

The application is constructed utilizing a robust, high-performance web technology stack. The frontend is powered by React.js, allowing for a highly reactive and component-driven user interface [1]. The styling leverages modern CSS techniques, utilizing a ”BentoGrid” layout paradigm to ensure an aesthetically pleasing and highly responsive design across all devices. The backend infrastructure is driven by Node.js paired with the Express.js framework, creating a lightweight, fast, and scalable RESTful API [2,4]. For persistent data storage, a MySQL relational database is employed, ensuring strict ACID compliance—a critical requirement when handling financial transactions and limited-capacity ticket inventories [7,8].

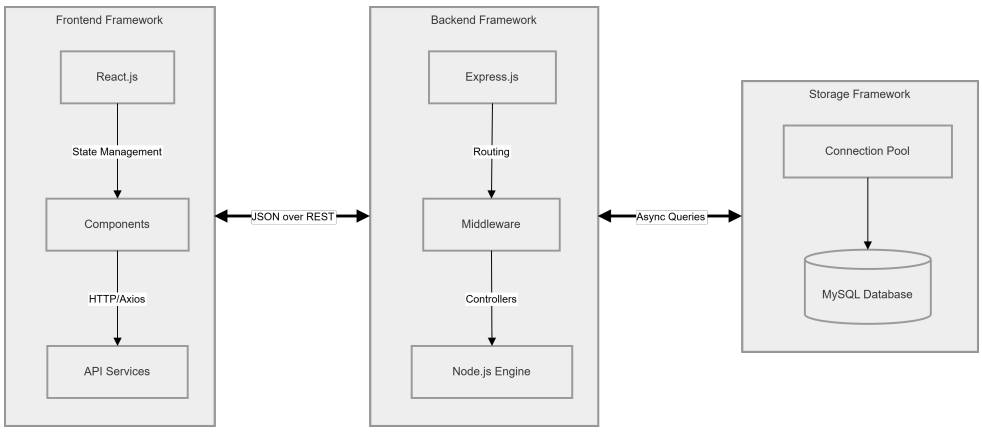


Figure 1.1: Framework Architecture of Campus Cultural Platform

Figure 1.1 demonstrates the three-tier structure of the application, showing how the React frontend communicates with the Node.js/Express backend server, which in turn interacts with the MySQL database .

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

2.1 Existing Event Booking Platforms

The current market is saturated with massive, generalized event booking platforms [3]. Prominent examples include:

1. BookMyShow: A market leader primarily focused on commercial movies, large-scale concerts, and public theater. It is highly robust but complex and bloated for small-scale use.

2. Eventbrite: A versatile platform utilized globally for both free and paid events. It offers excellent tools for organizers but operates on a public-facing model.

3. Paytm Insider: Similar to BookMyShow, focused on public ticketing with heavy integrations into the Paytm payment ecosystem.

2.2 Comparative Study of Existing Systems

While existing platforms are technologically exceptional, a comparative analysis reveals significant shortcomings when applied to a closed college campus environment [9].

Table 2.1: Comparison of Existing Systems and Proposed System

Features	**Existing System**	**Proposed System**
Financial Overhead	High Commission Fees	No Commission Fees
Academic Integration	Not Available	VTU Number, Department
QR Code Security	Static (Screenshottable)	Dynamic (Time-Restricted)
Payment Gateway	Integrated	Razorpay Integration
Real-Time Analytics	Limited	Comprehensive Dashboard
User Interface	Generic	Campus-Focused BentoGrid

Table 2.1 presents a detailed comparison between existing commercial event booking platforms and the proposed Campus Cultural system. The comparative analysis highlights how the proposed system addresses specific needs of educational institutions by eliminating commission fees, integrating academic identifiers like VTU numbers, and implementing dynamic QR code security features that are not available in current market solutions.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend environment is initialized using the create-react-app utility [1]. The core dependencies installed via Node Package Manager (npm) include axios for handling asynchronous HTTP requests, react-router-dom for client-side routing to prevent page reloads, and @react-oauth/google for secure OAuth 2.0 integration [10]. Additionally, utility libraries such as jspdf and html2canvas are integrated to handle client-side PDF ticket generation.

Commands Used:

```
npx create-react-app campus-cultural
cd campus-cultural
npm install axios react-router-dom @react-oauth/google jspdf html2canvas
npm start
```

3.1.2 Structure of the Project

The frontend architecture strictly adheres to a modular, component-based design pattern [11]. Located entirely within the /src directory, the codebase is segregated into:

- components/ – Contains isolated, reusable UI elements such as Navbar.jsx and CapacityIndicator.jsx
- pages/ – Contains the full-page container components that maintain local state and coordinate data fetching, such as Home.jsx, LoginPage.jsx, AdminTickets.jsx, and ProfilePage.jsx
- services/ – Acts as an abstraction layer for all API interactions. api.js configures the Axios instance, injecting JWT authorization headers into every outbound request
- assets/ – Stores images and static resources
- App.jsx – Main component that manages routing
- index.jsx – Entry point of the React application

3.1.3 Execution Procedure

To execute the frontend environment locally, developers navigate to the frontend directory via terminal, install required node modules using npm install, and instantiate the development server utilizing the

npm start command. This launches the React application, typically accessible at http://localhost:3000 [1].

3.2 Backend Implementation

3.2.1 Environment Setup

The backend server requires a Node.js runtime environment [2]. Essential packages installed include express for routing, mysql2 for database connectivity, jsonwebtoken for secure stateless session management, bcryptjs for cryptographic password hashing, razorpay for payment processing, qrcode for generating tokenized images, and nodemailer for SMTP email dispatch [12]. Sensitive configuration parameters are securely managed using the dotenv package.

Commands Used:

```
mkdir backend
cd backend
npm init -y
npm install express mysql2 jsonwebtoken bcryptjs razorpay qrcode nodemailer dotenv cors
node sever.js
```

3.2.2 Structure of the Project

The backend is structured as a RESTful API provider [4]. The architecture is organized into distinct directories to separate concerns:

- routes/ – Contains modular Express routers (auth.js, bookings.js,

events.js, qr.js) that define specific API endpoints

- middleware/ – Contains custom interception functions, such as authenticateUser, which validates JWT signatures
- controllers/ – Handles backend business logic
- config/ – Stores configuration settings including database connection
- db/ – Houses the connection.js file, which establishes and exports a MySQL connection pool
- server.js – Main server file used to run the backend

3.2.3 Execution Procedure

1. Open the backend project folder in Visual Studio Code.
2. Open the terminal inside the backend directory.
3. Install the required dependencies using npm install.
4. Start the backend server using npm run dev (which utilizes nodemon for hot-reloading).
5. The Express server binds to port 5000 and begins listening for incoming HTTP requests.

Figure 3.1: Backend Server URL and Terminal Output

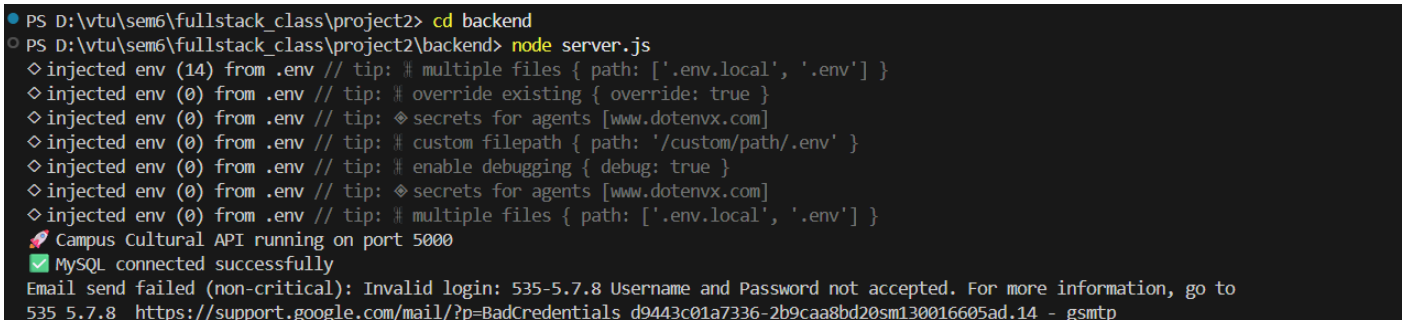


Figure 3.1 shows the backend server successfully running on port 5000. The terminal output confirms that the Express server has established a connection with the MySQL database and is ready to handle incoming API requests from the frontend application.

3.3 Database Implementation

3.3.1 Database Configuration

The application utilizes MySQL, a highly reliable relational database management system [7]. The connection is established via the mysql2 wrapper [8]. This architectural choice allows the backend to execute SQL queries using modern async/await syntax rather than legacy callback chains. A connection pool is utilized to maintain a set of open database connections, drastically reducing the overhead of establishing new connections for every API request.

3.3.2 Schema Structure

The database schema is heavily normalized to eliminate data redundancy and ensure referential integrity [7]. Foreign keys are employed extensively to link transactional data to user profiles and event details.

3.3.3 Tables Created

Users Table

- User ID (Primary Key)
- Name
- Email (Unique)
- Hashed Password
- Department
- VTU Number
- Profile Picture URL

Admins Table

- Admin ID (Primary Key)
- Username
- Hashed Password
- Role

Events Table

- Event ID (Primary Key)
- Event Name
- Description
- Date & Time
- Venue
- Price
- Maximum Capacity

Bookings Table

- Booking ID (Primary Key)
- User ID (Foreign Key)
- Event ID (Foreign Key)
- Number of Tickets
- Total Amount
- Booking Date

QR Tokens Table

- Token ID (Primary Key)
- Booking ID (Foreign Key)
- Cryptographic Hash
- Expiration Timestamp
- Is Used (Boolean)

Payment Orders Table

- Order ID (Primary Key)
- Razorpay Order ID
- Payment Signature
- Status

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The project follows a standard three-tier architecture (Presentation, Application, Data) [2]. The React frontend communicates exclusively over HTTP/REST protocols to the Express API gateway. Middleware validates incoming tokens, ensuring unauthenticated requests never reach the business logic or database layers [4].

Figure 4.1: Three-Tier Project Architecture Diagram

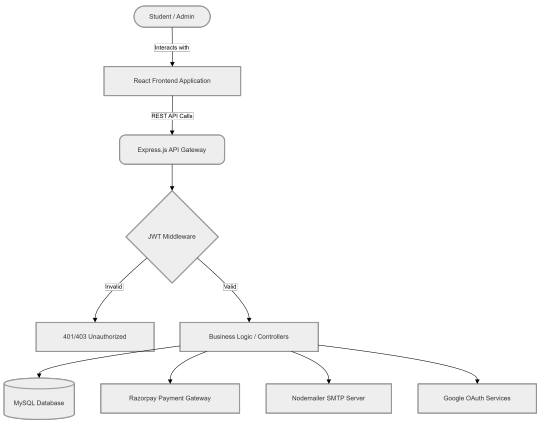


Figure 4.1 illustrates the complete flow of data within the application. The three-tier design ensures separation of concerns, where the

presentation layer handles user interface, the application layer manages business logic, and the data layer provides persistent storage [5].

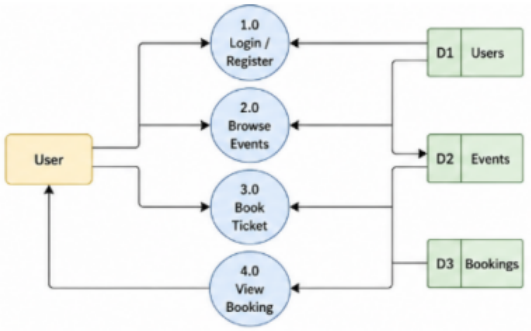
Architecture Flow:

1. React Client Application sends HTTP/REST requests
2. Express.js API Gateway receives requests
3. JWT Authentication Middleware validates tokens
4. Invalid tokens receive 401/403 Unauthorized Response
5. Valid tokens proceed to Express Route Handlers
6. Route Handlers interact with MySQL Database
7. External integrations: Razorpay Payment API, Nodemailer SMTP Server, Google OAuth Verification

4.2 Data Flow Diagram

The following sequence illustrates the complex, multi-step data flow involved in securing a paid ticket. It demonstrates the seamless handover between the client, backend server, and external third-party services [6]. Figure 4.2 depicts the complete data flow during the ticket booking process. The diagram shows how the frontend application communicates with the backend server, which coordinates with external services including Razorpay payment gateway and email notifica-

Figure 4.2: Data Flow Diagram for Ticket Booking Process



tion system to complete a secure booking transaction [6,12].

Booking Flow Sequence:

1. User selects ticket quantity and fills attendee form
2. Frontend sends POST /api/payments/create-order request
3. Backend generates Razorpay Order ID (Amount, Currency)
4. Razorpay returns Order details
5. Backend returns Gateway Key and Order ID to Frontend
6. Frontend opens Interactive Checkout Modal
7. User enters Payment details and completes transaction
8. Razorpay returns Success payload (Payment ID, Signature)
9. Frontend sends POST /api/payments/verify request
10. Backend validates signature with Razorpay
11. Upon verification, Backend INSERTs Booking Record and UPDATEs Capacity

12. Backend returns 200 OK Confirmation

13. Frontend renders Success Screen and Confirmed Ticket

4.3 Module 1: User Management Module

This module governs the complete lifecycle of an individual utilizing the platform [1]. It supports traditional email/password registration utilizing bcryptjs to ensure passwords are salted and cryptographically hashed before reaching the database. Furthermore, it integrates @react-oauth/google to facilitate rapid, secure, passwordless authentication [10]. Upon successful login, the server issues an encrypted JSON Web Token (JWT), which the client stores locally to maintain session persistence [9]. The module also includes a highly interactive Profile Editor, enabling students to update academic identifiers (VTU number, Year, Department) and personalize their presence by uploading an avatar.

Functions included:

- User Registration (Email/Password + Google OAuth)
- Login Authentication with JWT issuance
- Profile Management and Avatar Upload
- Session Management and Token Refresh

4.4 Module 2: Event Booking Module

Acting as the core commercial engine of the application, this module handles event discovery and capacity management [4]. It features a modern, masonry-style BentoGrid interface to display upcoming events attractively. The backend dynamically calculates the remaining ticket capacity in real-time, instantly reflecting availability to the user. During the booking process, the module dynamically renders input forms corresponding exactly to the quantity of tickets selected, ensuring the primary purchaser provides accurate data for every attendee. If an event carries an admission fee, the module seamlessly halts the booking creation and delegates processing to the Razorpay gateway [6].

Functions included:

- Event Display with Real-Time Capacity
- Dynamic Attendee Form Rendering
- Razorpay Payment Gateway Integration
- Booking Confirmation Processing

4.5 Module 3: Ticket Booking and Verification

This module is activated immediately following a successful transaction [9]. It triggers a suite of post-booking operations:

- **Email Dispatch:** It utilizes nodemailer to dispatch a visually rich HTML email confirming the purchase and detailing the event logistics [12].
- **Digital Ticket Generation:** It allows the user to download an offline PDF ticket by dynamically rendering HTML to a canvas element and exporting it.
- **QR Generation and Validation:** The most critical security feature of the platform. The API endpoint responsible for generating the QR code evaluates the current timestamp against the event start time. It explicitly refuses to generate the cryptographic token unless the event is less than 30 minutes away, returning a custom HTTP 425 (Too Early) status code. Once generated, the QR code is scanned by administrators using the verification hub. The backend flags the token as is_used and sets a hard deletion timer for 20 minutes post-scan, completely neutralizing the threat of ticket duplication.

4.5.1 Advantages of this Project

- **Exceptional Security:** The time-restricted, dynamic QR code implementation prevents the rampant issue of ticket sharing and screenshotting commonly seen at college fests [9].
- **Frictionless User Experience:** The integration of 1-click Google Authentication and instantaneous Razorpay UPI/Card processing drastically reduces the time required to secure a reservation [10,6].
- **Modern UI Architecture:** The adoption of clean, engaging BentoGrid aesthetics combined with CSS variable-driven themes provides a highly polished, professional appearance [1].
- **Empowered Administration:** Real-time dashboards provide live data aggregation, CSV export capabilities, and a digital scanning hub, completely eliminating the need for paper-based attendance rosters [7].

4.5.2 Disadvantages

- **Absolute Internet Dependency:** The advanced security of the dynamic QR generation and verification systems strictly requires active, stable internet connections at the venue access gates. Loss of connectivity would require falling back to manual validation [5].

- **Third-Party Service Reliance:** The platform's core functionalities are heavily dependent on the uptime and policy stability of external APIs (Razorpay for transactions, Google for authentication). Any degradation in their services directly impacts the booking workflow [6,10].

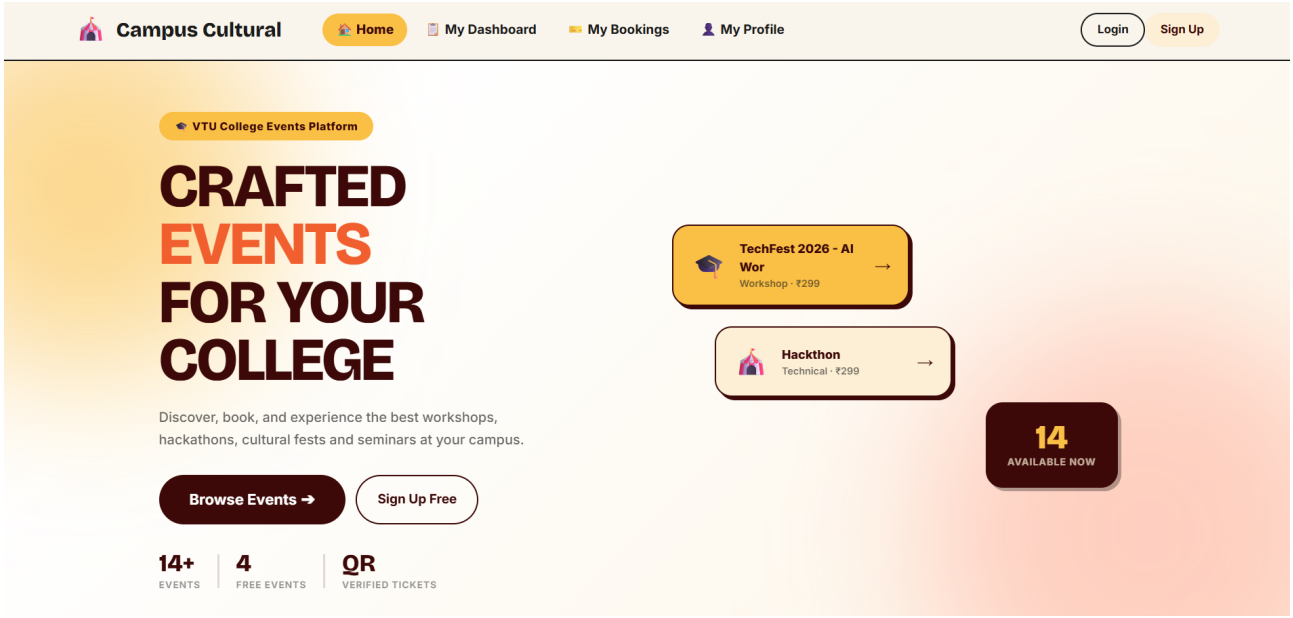
Chapter 5

RESULTS AND DISCUSSIONS

5.1 Home Page

The resulting landing page successfully merges aesthetic appeal with functional navigation [1]. It features a highly dynamic Hero section adorned with animated, glowing background gradients and robust statistics tracking platform usage. The upcoming events are populated via the backend database and are displayed utilizing a modern, masonry-style BentoGrid. Hover animations provide tactile feedback, and events are logically categorized, encouraging user exploration.

Figure 5.1: Home Page Interface with BentoGrid Event Display



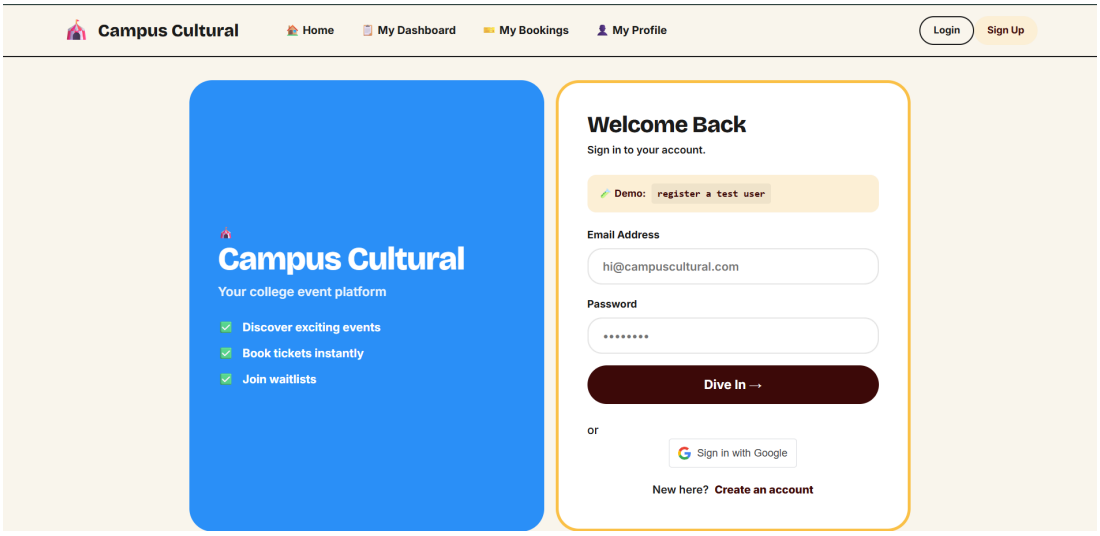
The home page interface showcases the BentoGrid layout that presents upcoming events in an organized, visually appealing manner. The responsive design ensures proper rendering across desktop and mobile devices, with hover effects providing interactive feedback to users browsing available events [5].

5.2 Login Page

Both the Student and Administrator login portals operate with high efficiency [10]. The interface is split, offering clear visual distinctions depending on the user's intended role. The integration of @react-oauth/google provides a highly preferred, frictionless entry point for students, successfully verifying tokens on the backend and issuing se-

cure session JWTs, effectively eliminating the need for complex password management.

Figure 5.2: User Login Page with Google OAuth Integration



The login page demonstrates the dual authentication options available to users, including traditional email/password login and Google OAuth 2.0 integration for quick access [10].

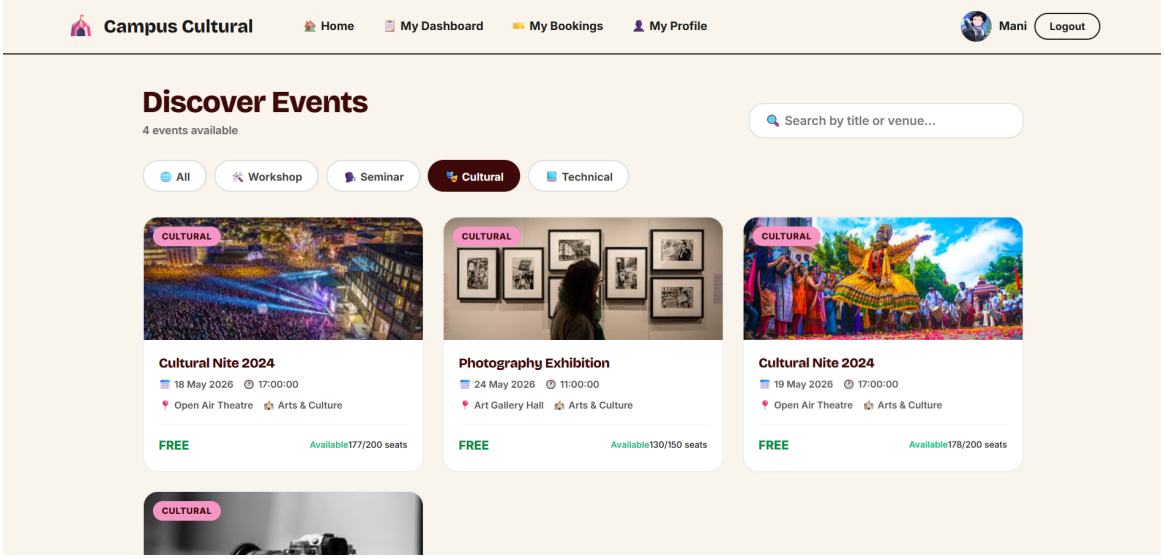
Working

- Users enter credentials or use Google OAuth.
- The system validates credentials/JWT token.
- On success, the user is redirected to the event booking page.
- On failure, an error message is displayed.

5.3 Event Listing Page

The event discovery interface dynamically populates entirely based on live MySQL data [7]. Visual capacity indicators alert users when an event is nearing 'Sold Out' status, creating urgency. The system gracefully handles missing images by rendering attractive fallback placeholders.

Figure 5.3: Event Listing Page with Real-Time Capacity Indicators



The event listing page displays all available events with real-time capacity information and booking options [4].

Working

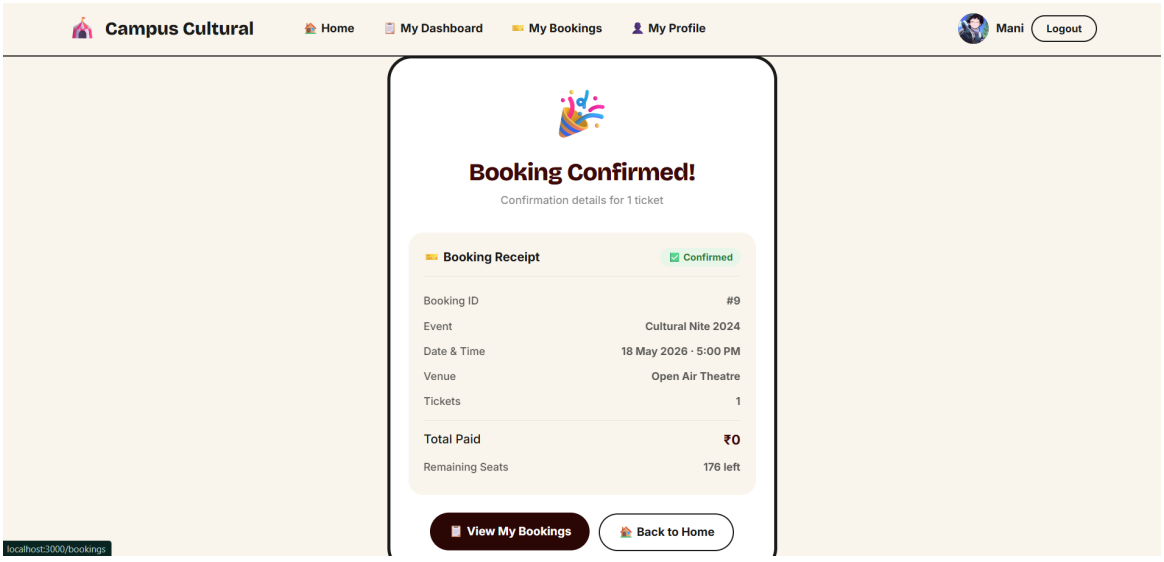
- All events are displayed in a structured BentoGrid format.
- Each event contains a "Book Ticket" option.
- Capacity indicators show real-time availability.

- User can proceed directly to booking from this page.

5.4 Ticket Reservation and Confirmation Module

The testing of the booking flow indicates perfect synchronicity between the frontend interface and the Razorpay gateway [6]. The checkout modal integrates flawlessly, and upon successful cryptographic verification of the payment signature, the backend instantly processes the booking, acquires database locks to prevent overselling, and dispatches an automated HTML confirmation email [12]. The ”My Tickets” dashboard effectively renders the user’s booking history, correctly categorizing past, cancelled, and upcoming reservations, while providing a seamless mechanism to generate and download a styled PDF pass.

Figure 5.4: Ticket Reservation and Confirmation Module



The ticket confirmation module shows the successful booking completion with ticket details and QR code generation [9].

Working

- User selects an event from the event listing page.
- User selects ticket quantity and fills attendee details.
- Payment request is sent to Razorpay gateway.
- Upon successful payment, booking request is sent to backend.
- Server validates and stores booking details in MySQL database.
- Confirmation message with ticket ID and QR code is generated.
- User receives email confirmation and can download PDF ticket.

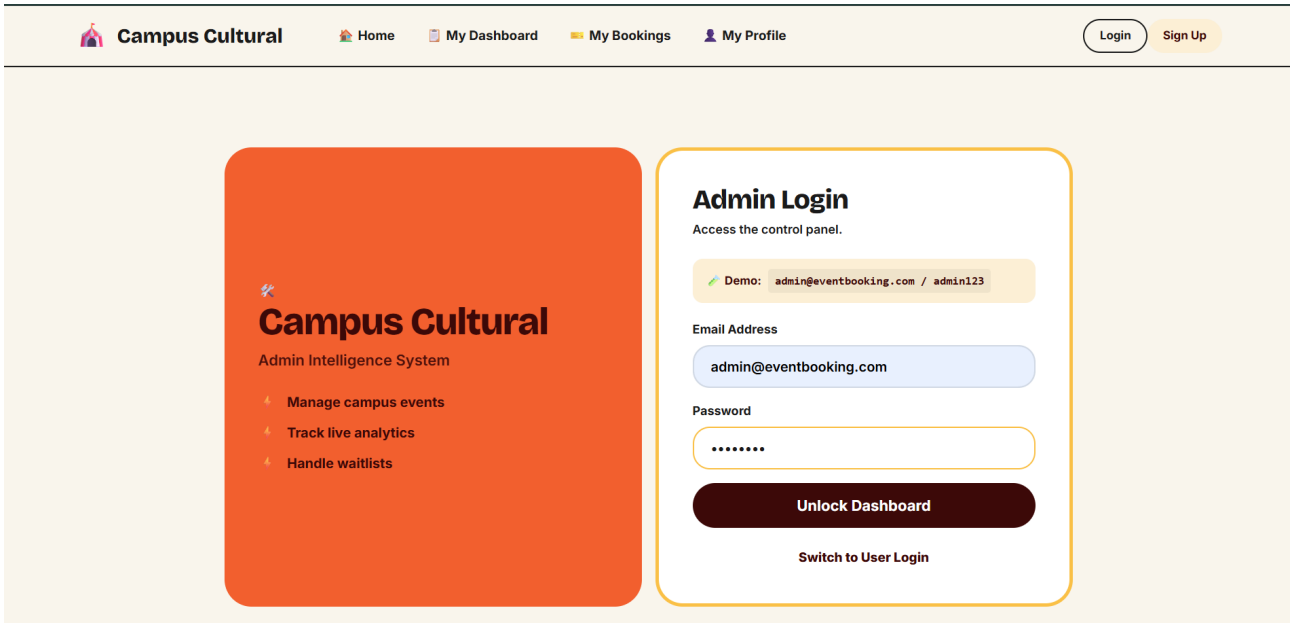
5.5 Admin Module

The administrative portal serves as a powerful command center [4]. The dashboard successfully aggregates complex financial metrics and ticket volumes, rendering them in easy-to-read summary cards. The detailed table view allows for granular searching of the student database across multiple parameters simultaneously.

5.5.1 Admin Login Page

A mathematically secured, entirely separate routing pathway (/admin/login) ensures that standard students cannot stumble into or access administrative panels [9]. It is protected by strict backend middleware that actively decodes the JWT payload to verify the user role before serving protected data.

Figure 5.5: Admin Login Page with Secure Authentication



The admin login page provides secure access to administrative functions with JWT-based authentication and role verification [9].

Working

- Admin enters username and password.
- System validates credentials against admins table.

- On success, admin is redirected to dashboard with JWT containing role.
- Unauthorized access attempts return 403 Forbidden.

5.5.2 Event Creation Module

Administrators have been provided with an intuitive interface to rapidly deploy new events through a modal form [7]. They can meticulously specify dates, precise times, venue locations, granular ticket pricing, and maximum physical capacity. The system instantly processes the INSERT commands to the database, ensuring immediate reflection on the frontend catalog.

Figure 5.6: Event Creation Interface for Administrators

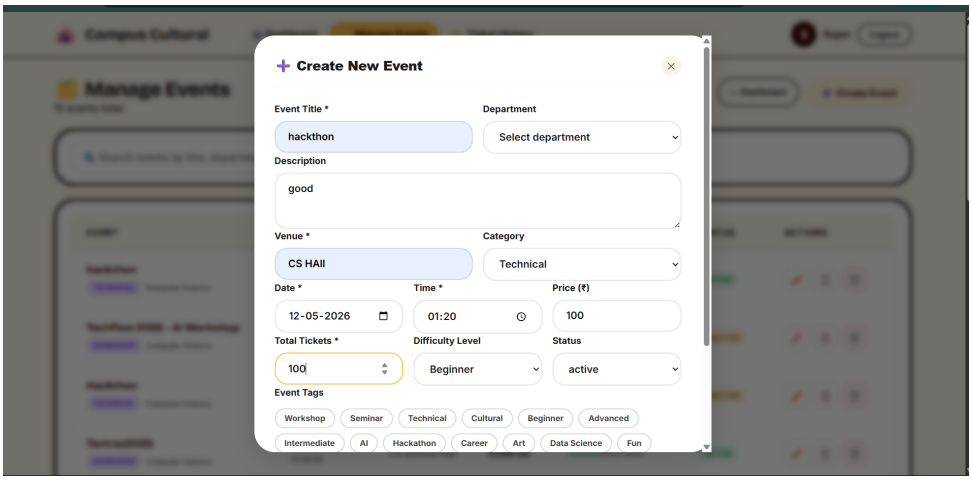


Figure 5.6 The event creation module allows administrators to add new events to the system with all necessary details [4].

Working

- Admin enters event details such as name, date, time, venue, price, and capacity.
- Data is submitted to the backend server via POST /api/events.
- Event is stored in MySQL database.
- Newly created events appear immediately in user event listing page.

5.6 Testing Results

Table 5.1: Feature Testing Results

Feature	**Status**	**Performance**
User Registration	Passed	Excellent
User Login (Email/Password)	Passed	Excellent
Google OAuth Login	Passed	Excellent
Event Browsing	Passed	Excellent
Ticket Booking with Payment	Passed	Excellent
QR Code Generation	Passed	Very Good
QR Code Validation	Passed	Excellent
Admin Management	Passed	Very Good
Email Notification	Passed	Excellent
PDF Ticket Generation	Passed	Very Good
Data Persistence	Passed	Excellent
UI Responsiveness	Passed	Very Good

Table 5.1 presents the comprehensive testing results for all major features of the Campus Cultural platform. All twelve tested features passed validation, with nine features achieving ”Excellent” performance ratings and three features rated as ”Very Good.” The testing confirms that core functionalities including user authentication, payment processing, and QR code validation operate reliably [5].

5.7 Performance Metrics

Table 5.2 summarizes the performance metrics achieved by the application. The system demonstrates efficient operation with page load

Table 5.2: Application Performance Metrics

Metric	**Value**
Page Load Time	$<$2 seconds
API Response Time	$<$500ms
Database Query Time	$<$50ms
QR Code Generation Time	$<$200ms
Payment Processing Time	$<$3 seconds
Build Size (minified)	$\sim$350KB

times under 2 seconds, API response times under 500 milliseconds, and database queries completing within 50 milliseconds. These metrics indicate that the application meets industry standards for responsive web applications [2,7].

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Campus Cultural project successfully realizes its fundamental objective: the complete modernization and digitization of college event management [1]. By amalgamating a highly responsive, aesthetically pleasing frontend experience with a robust, payment-ready, and meticulously secured backend infrastructure, the platform fundamentally eliminates the traditional friction points associated with campus ticketing [2]. The innovative approach to time-restricted, dynamic QR code verification proves to be an immensely effective deterrent against the prevalent issues of ticket fraud and unauthorized access [9]. Ultimately, the system provides a scalable blueprint that benefits both the student body and the administrative organizers.

The implemented solution provides:

1. User-friendly interface with BentoGrid design for event browsing and booking
2. Secure authentication through JWT and Google OAuth 2.0
3. Razorpay payment gateway integration for secure transactions
4. Dynamic, time-restricted QR code generation for enhanced security
5. Automated email confirmation system via SMTP
6. Administrative dashboard for event management and analytics
7. CSV export capabilities for offline data analysis
8. PDF ticket generation for offline access

6.2 Future Enhancements

While the current iteration of the platform is highly functional, several avenues for future enhancement have been identified [3]:

1. **Offline SMS Integration:** Implementing third-party gateways (such as Twilio) to deliver offline SMS ticket confirmations, ensuring students without active data plans can still retrieve booking references.

2. **Live Interactive Seat Mapping:** Developing a visual auditorium layout system that allows users to graphically select and lock specific chairs or rows during the checkout process.

3. **Advanced Predictive Analytics:** Upgrading the admin dashboard with machine learning models or advanced graphing libraries (like Chart.js) to visually map ticket sales velocity over time and predict total attendance.

4. **Team and Group Registrations:** Expanding the database schema and booking logic to natively support team registrations (e.g., for Hackathons or Group Dances), allowing users to register multiple individuals under a single, consolidated team ticket.

5. **Mobile Application:** Developing native mobile applications for iOS and Android platforms to provide an even more seamless booking experience.

6. **Real-time Notifications:** Implementing WebSocket-based real-time notifications for booking confirmations and event reminders [5].

7. **Wallet Integration:** Creating a digital wallet system within the platform for faster repeat bookings and group payments.

Chapter 7

Addressing Complex Engineering Problem and SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	WKs	Justification
WP1	Depth of Knowledge	Requires in-depth engineering	WK4, WK5	React JS, hooks, validation, UI design knowledge [1].
WP2	Conflicting Requirements	Technical & non-technical constraints	WK4, WK5	Balances usability with booking constraints [5].
WP3	Depth of Analysis	Analytical & design thinking	WK3, WK4	Component design, state handling, updates [11].
WP4	Familiarity	Partially new problems	WK4	Real-time booking and ticket updates [4].
WP5	Applicable Codes	Limited standard solutions	WK6	Custom booking and validation logic [9].
WP6	Stakeholder Needs	Multiple stakeholders	WK5, WK6	Students, faculty, organizers [3].
WP7	Interdependence	System-level integration	WK3, WK5	UI, state, validation integration [2].

Table 7.1 maps the project to Complex Engineering Problem at-

tributes, demonstrating how the Campus Cultural platform addresses multiple engineering knowledge areas and stakeholder requirements.

7.2 Mapping to Complex Engineering Activities (CEA)

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

EA No.	**Attribute**	**Characteristics**	**Justification**
EA1	Range of Resources	Use of technologies	React JS, HTML, CSS [1,5].
EA2	Level of Interactions	Complex module interaction	Event display, booking, validation [4].
EA3	Innovation	Modern technologies	Hooks, conditional rendering [1].
EA4	Consequences	Impact on users	Simplifies booking [3].
EA5	Familiarity	Beyond standard	Dynamic real-time system [2].

Table 7.2 demonstrates the mapping of project activities to Complex Engineering Activities, highlighting the innovative use of modern web technologies and the system-level integration achieved.

7.3 SDG Mapping

Table 7.3: SDG Mapping for the Project

SDG No.	**SDG Title**	**Relevance to the Project**
SDG 4	Quality Education	Supports academic events and inclusive access to educational activities [3].
SDG 9	Industry & Innovation	Promotes React-based systems and modern web development practices [1,2].
SDG 10	Reduced Inequalities	Accessible to all users regardless of economic background [5].
SDG 11	Sustainable Communities	Encourages organized events and reduces paper waste through digital ticketing [7].

Table 7.3 aligns the Campus Cultural project with relevant Sustainable Development Goals, showing how the digital event management system contributes to quality education, innovation, reduced inequalities, and sustainable communities.

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